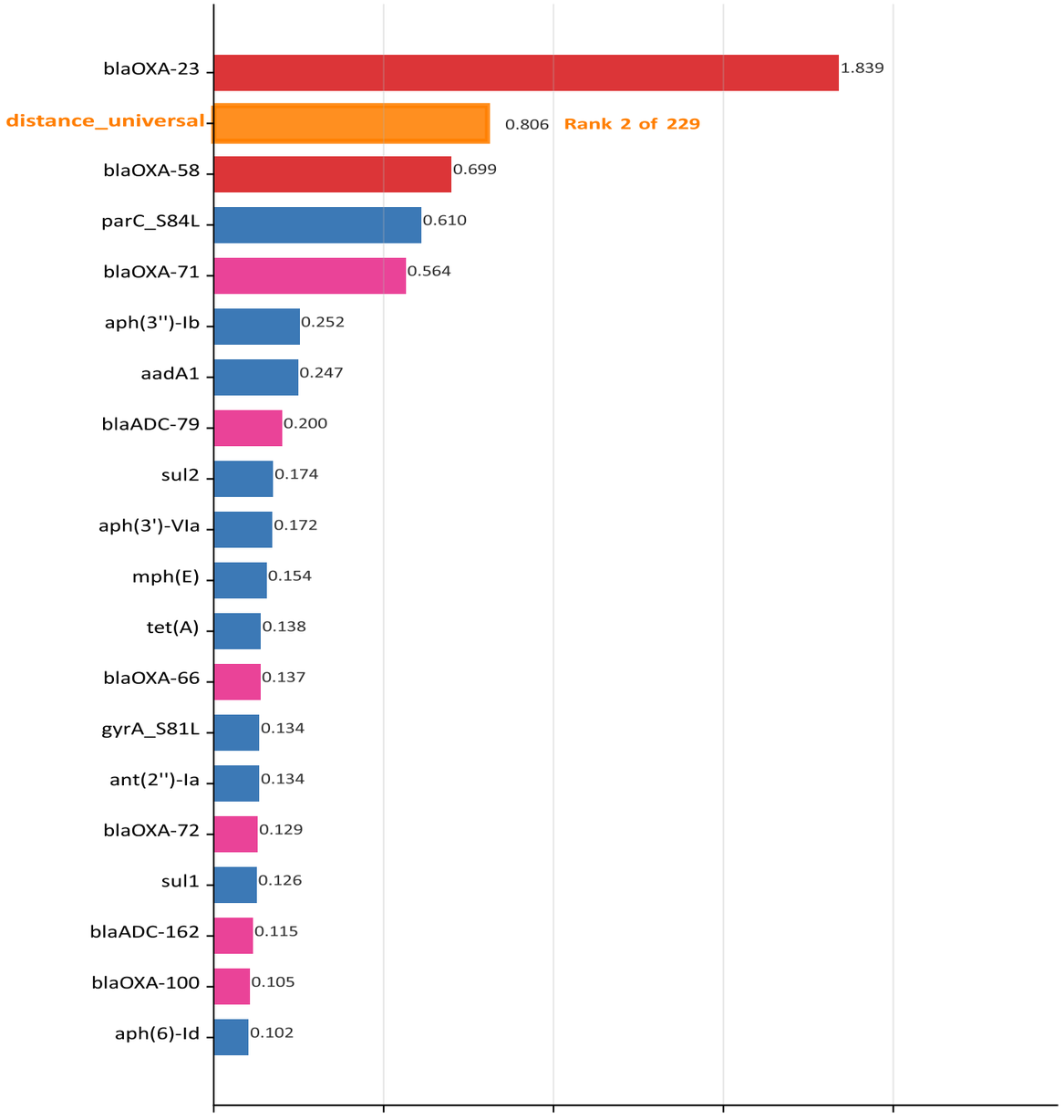
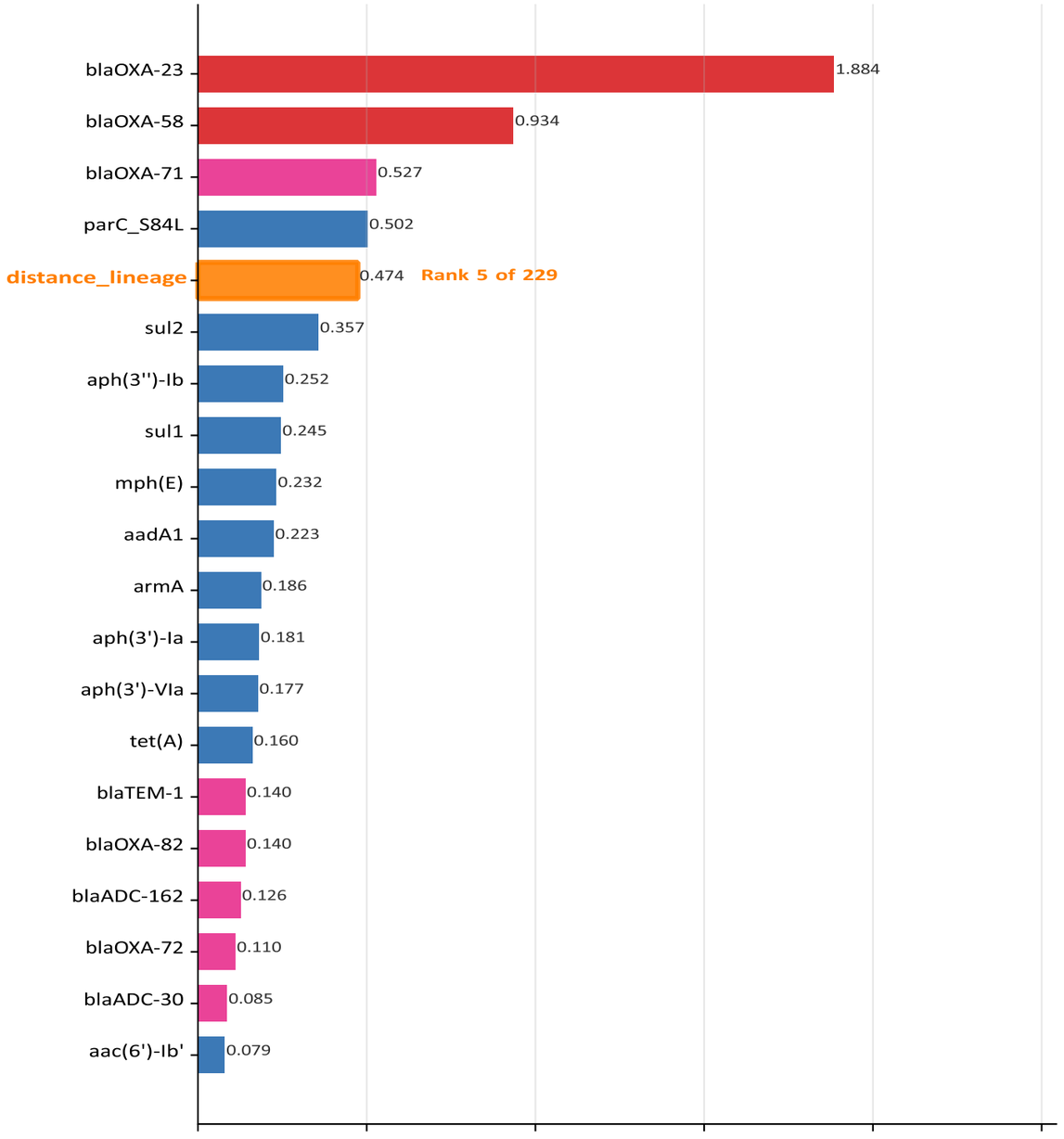
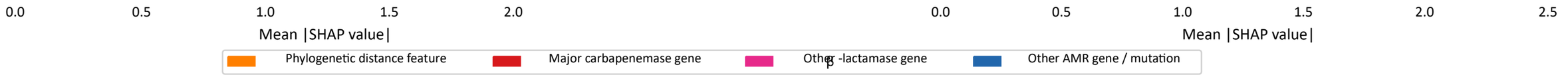
SHAP Feature Importance XGBoost Models
Top 20 features by mean absolute SHAP value

A Feature Set 4: Universal distance + AMR genes

B Feature Set 5: Lineage distance + AMR genes


blaOXA-23 1.839
distance_universal 0.806 Rank 2 of 229
blaOXA-58 0.699
parC_S84L 0.610
blaOXA-71 0.564
aph(3'')-Ib 0.252
aadA1 0.247
blaADC-79 0.200
sul2 0.174
aph(3')-VIa 0.172
mph(E) 0.154
tet(A) 0.138
blaOXA-66 0.137
gyrA_S81L 0.134
ant(2'')-Ia 0.134
blaOXA-72 0.129
sul1 0.126
blaADC-162 0.115
blaOXA-100 0.105
aph(6)-Id 0.102


blaOXA-23 1.884
blaOXA-58 0.934
blaOXA-71 0.527
parC_S84L 0.502
distance_lineage 0.474 Rank 5 of 229
sul2 0.357
aph(3'')-Ib 0.252
sul1 0.245
mph(E) 0.232
aadA1 0.223
armA 0.186
aph(3')-Ia 0.181
aph(3')-VIa 0.177
tet(A) 0.160
blaTEM-1 0.140
blaOXA-82 0.140
blaADC-162 0.126
blaOXA-72 0.110
blaADC-30 0.085
aac(6')-Ib' 0.079


0.0
0.5
1.0
1.5
2.0
Mean |SHAP value|
Phylogenetic distance feature
Major carbapenemase gene
Other -lactamase gene
Other AMR gene / mutation
0.0
0.5
1.0
1.5
2.0
2.5
Mean |SHAP value|